



EarStudio: Performance Evaluation

Single-Ended 3.5mm output w/ SONY XBA-N1 (16 Ω) earphones

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1. 24bit Bluetooth performance test

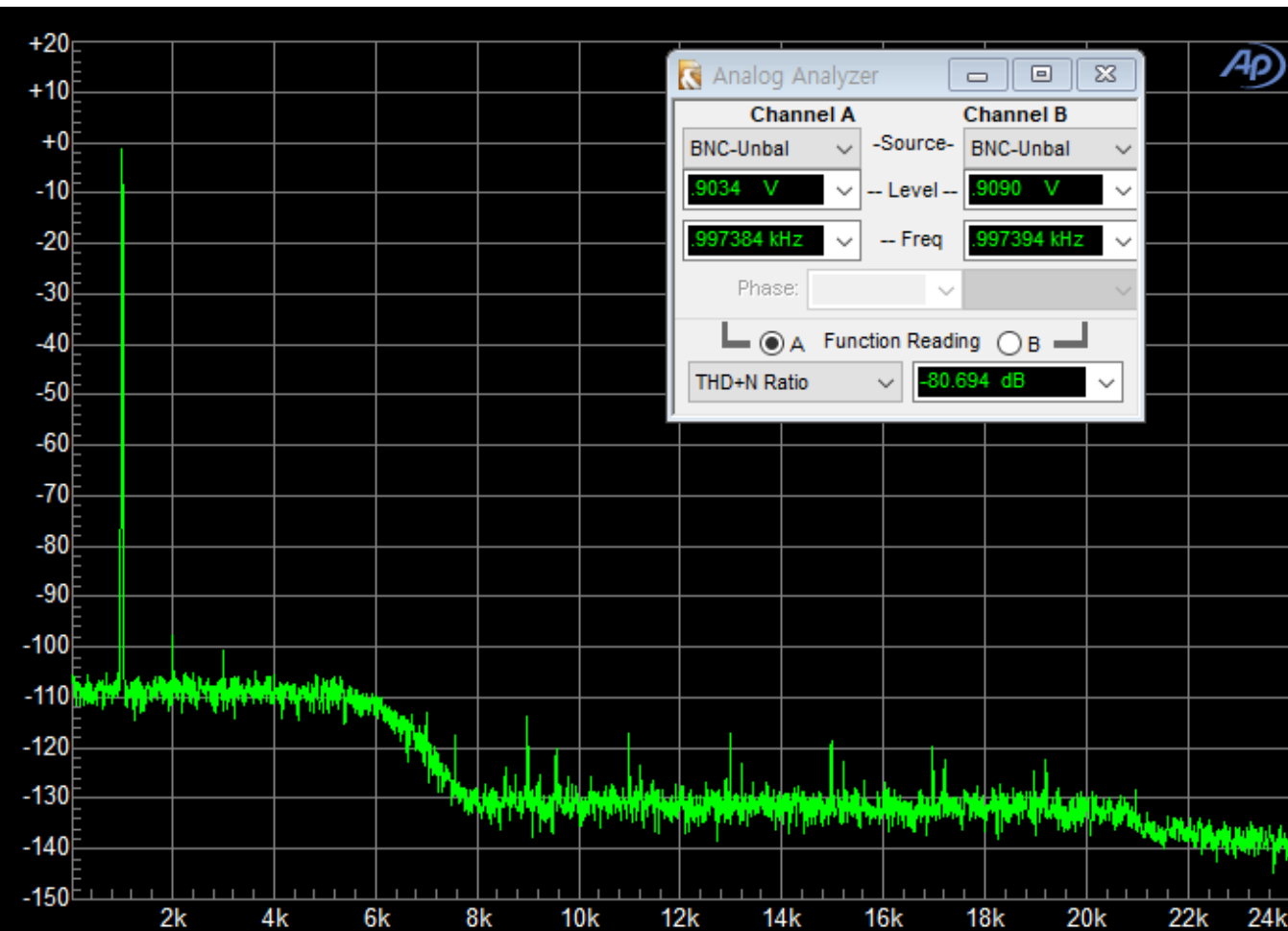
EarStudio vs. XB10

Test Setup:

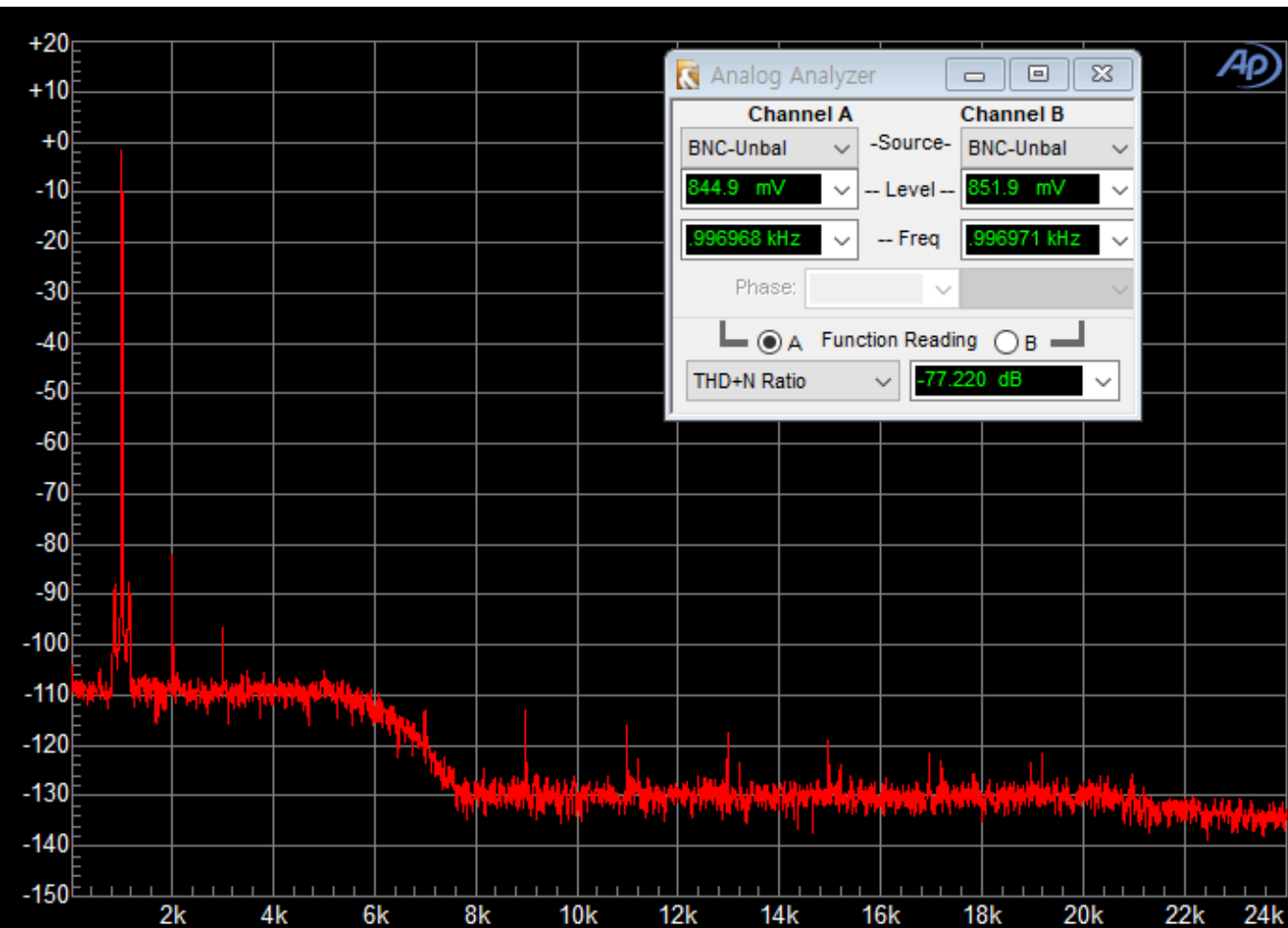
- Both EarStudio and XB10 work as an **aptXHD** receiver (i.e. Bluetooth Sink)
- AK70 (Bluetooth Source) streams the local test tone over A2DP
- 997KHz test tone
- SONY XBA-N1 (16Ω) connected as a load (Standard 3.5mm cable)
- Measurement Equipment: AP2722
- DUT 3.5mm single-ended output to AP2722 unbalanced input



No Load @ 840 mVrms



EarStudio (aptX-HD)



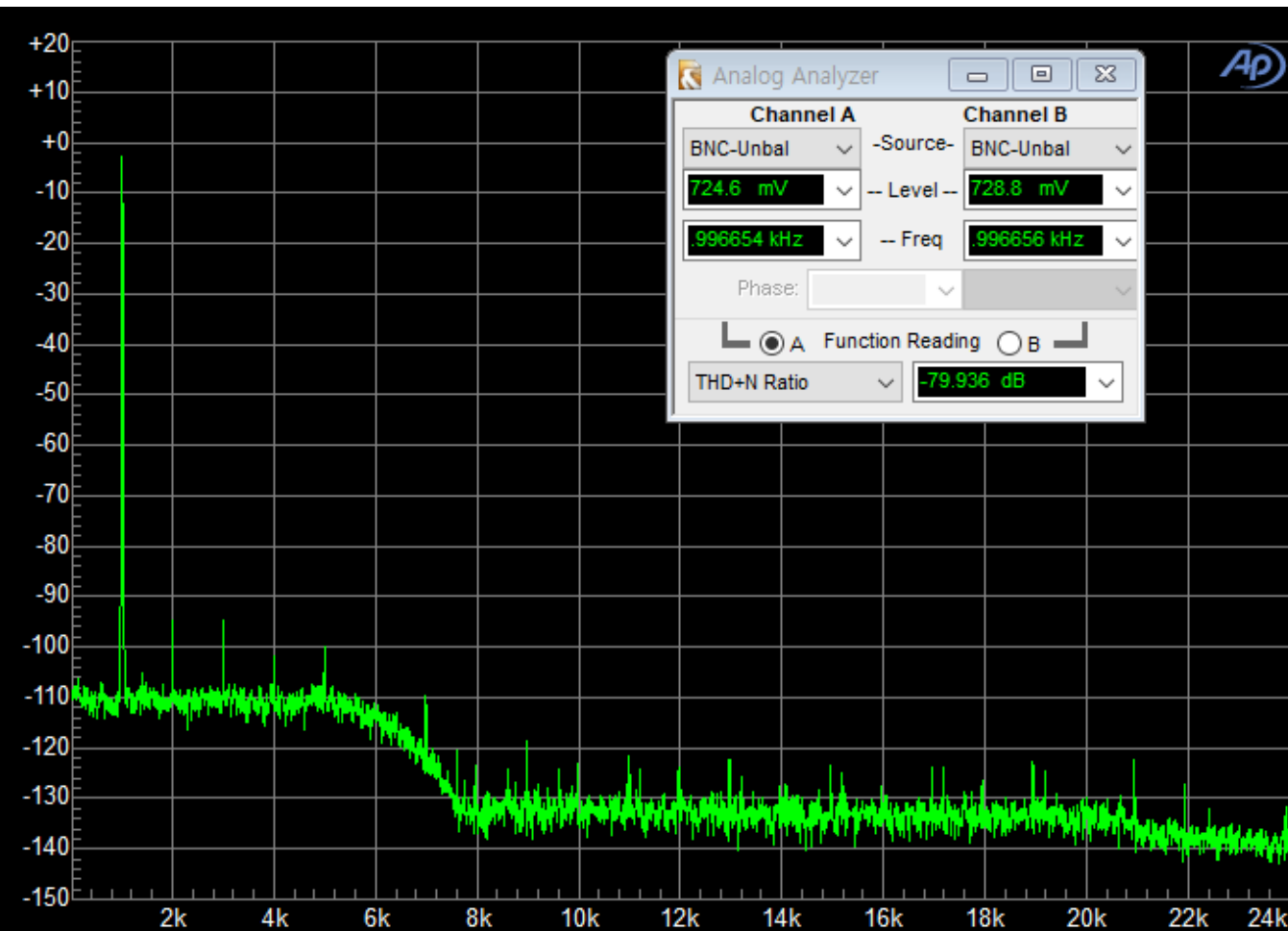
XB10 (aptX-HD)

Remark:

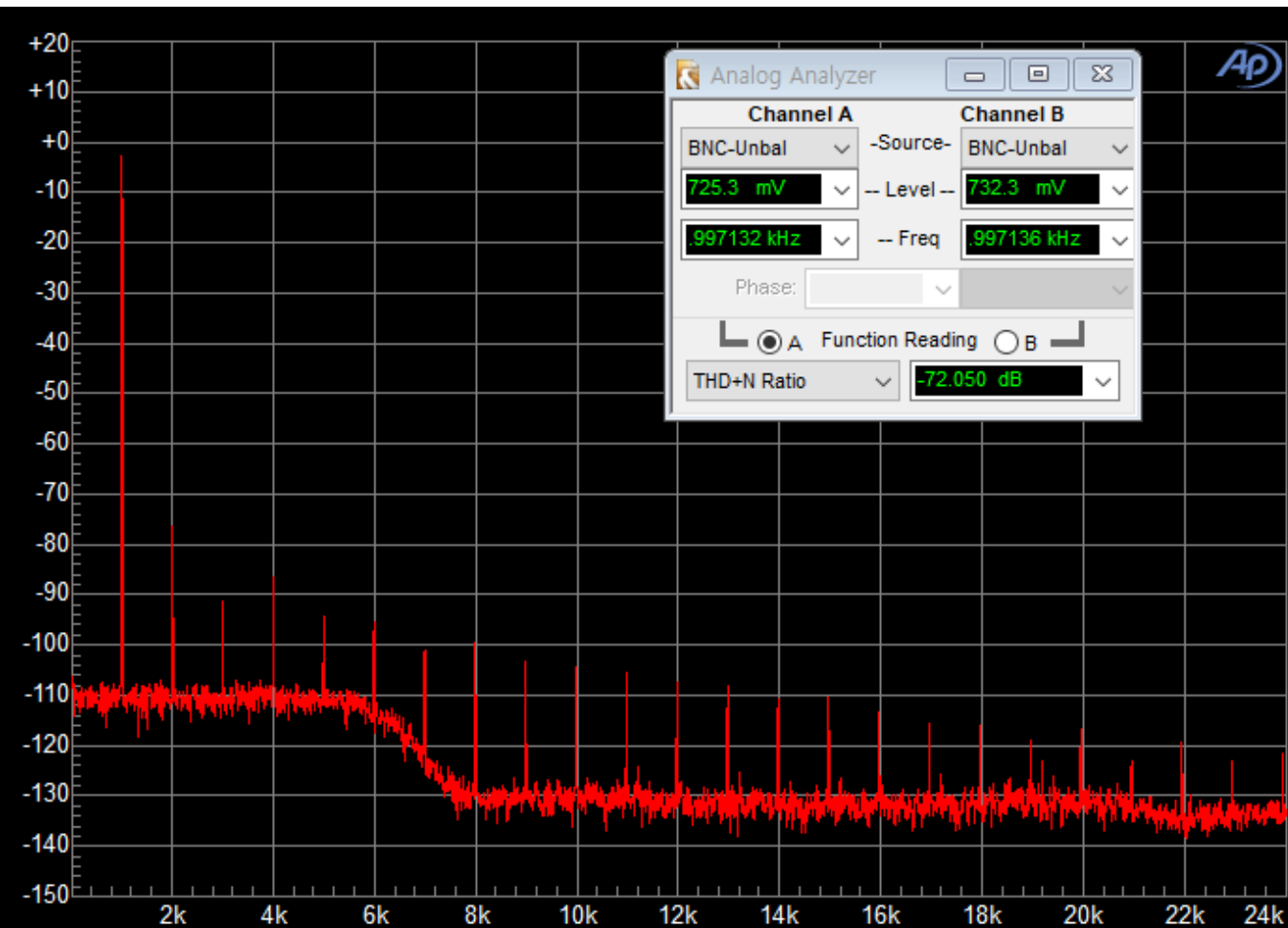
- When No load is connected, EarStudio (-80dB) is slightly better than XB10 (-77dB) regarding THD+N at 1.14Vrms output.
- The same Noise Floor

The digital domain THD+N of aptX-HD codec is limited to 80dB. The measurement results show the EarStudio analog front-end provides adequate performance which can thoroughly reproduce the source from the aptX-HD codec without any additional distortion or loss in the analog domain.

No Load @ 720 mVrms



EarStudio (aptX-HD)



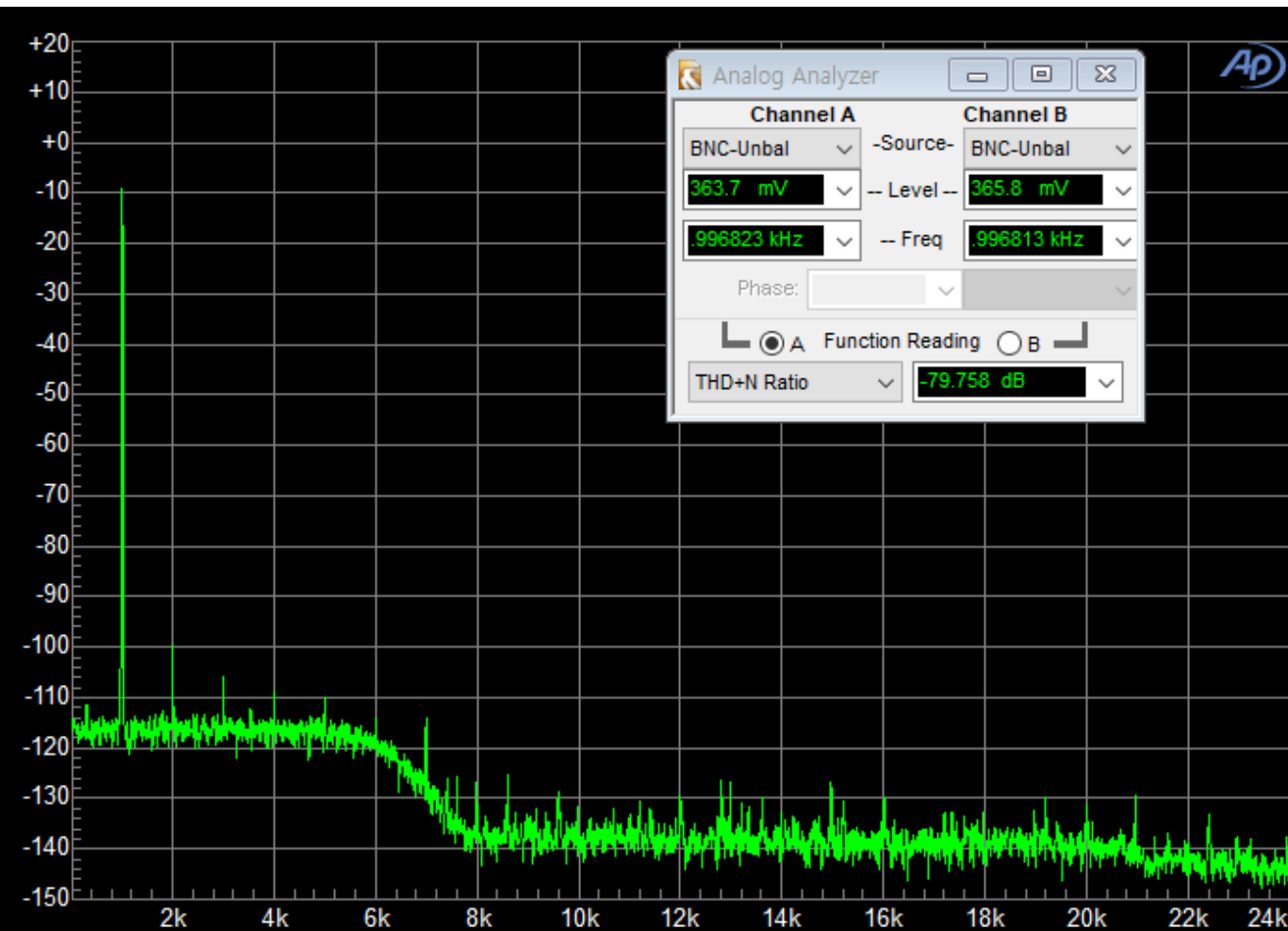
XB10 (aptX-HD)

Remark:

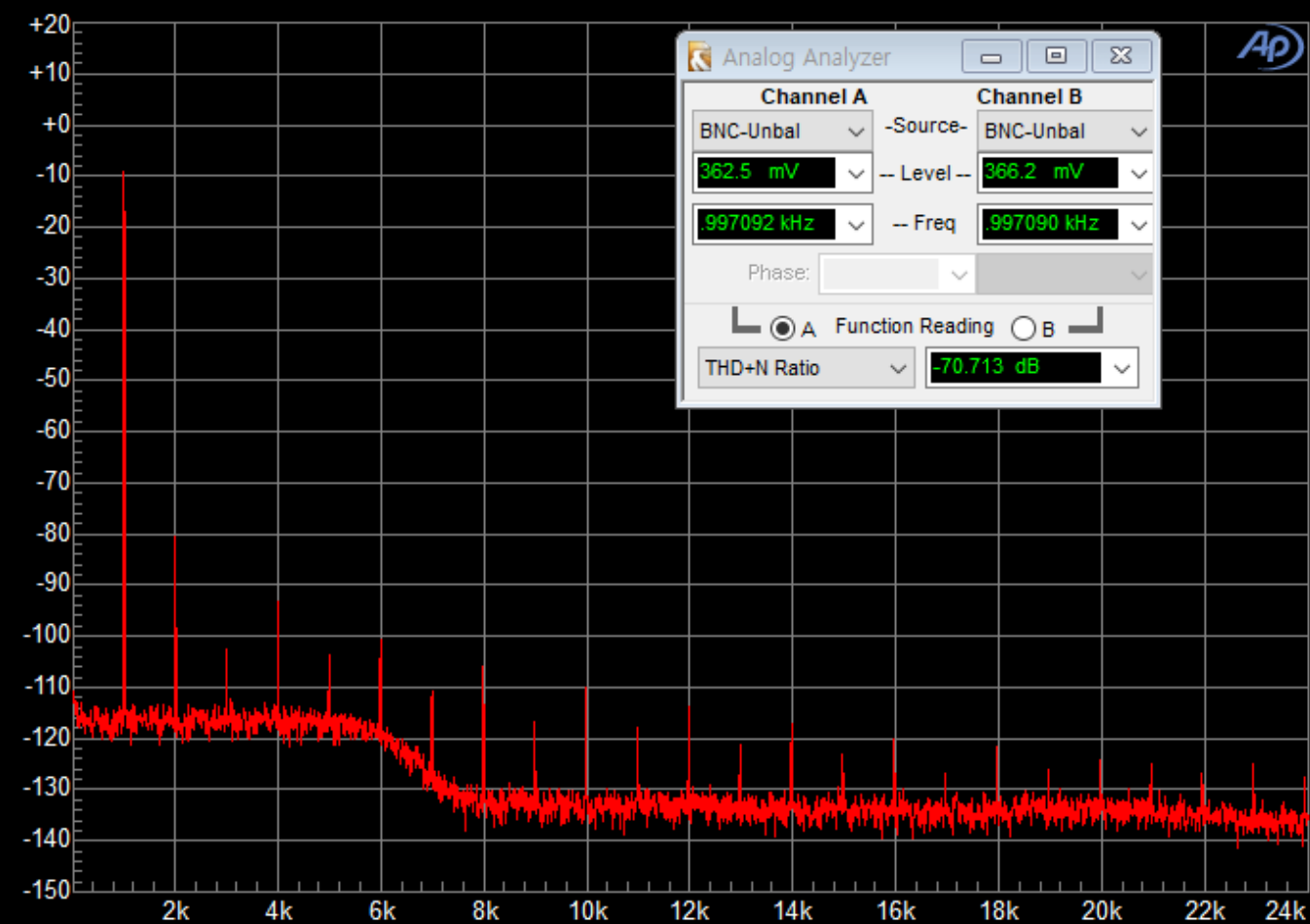
- When XBA-N1 (16 Ω) load is connected, EarStudio (-79dB) is better than XB10 (-72dB) regarding THD+N at 720 mVrms output.
- The Noise Floor of EarStudio is lower than that of XB10.

As mentioned earlier, driving 16 Ω earphones is not that easy. As shown above, XB10 has difficulty in driving the 16 Ω load, while EarStudio still provides enough power without any loss and degradation into the same load at the same output level.

No Load @ 360 mVrms



EarStudio (aptX-HD)



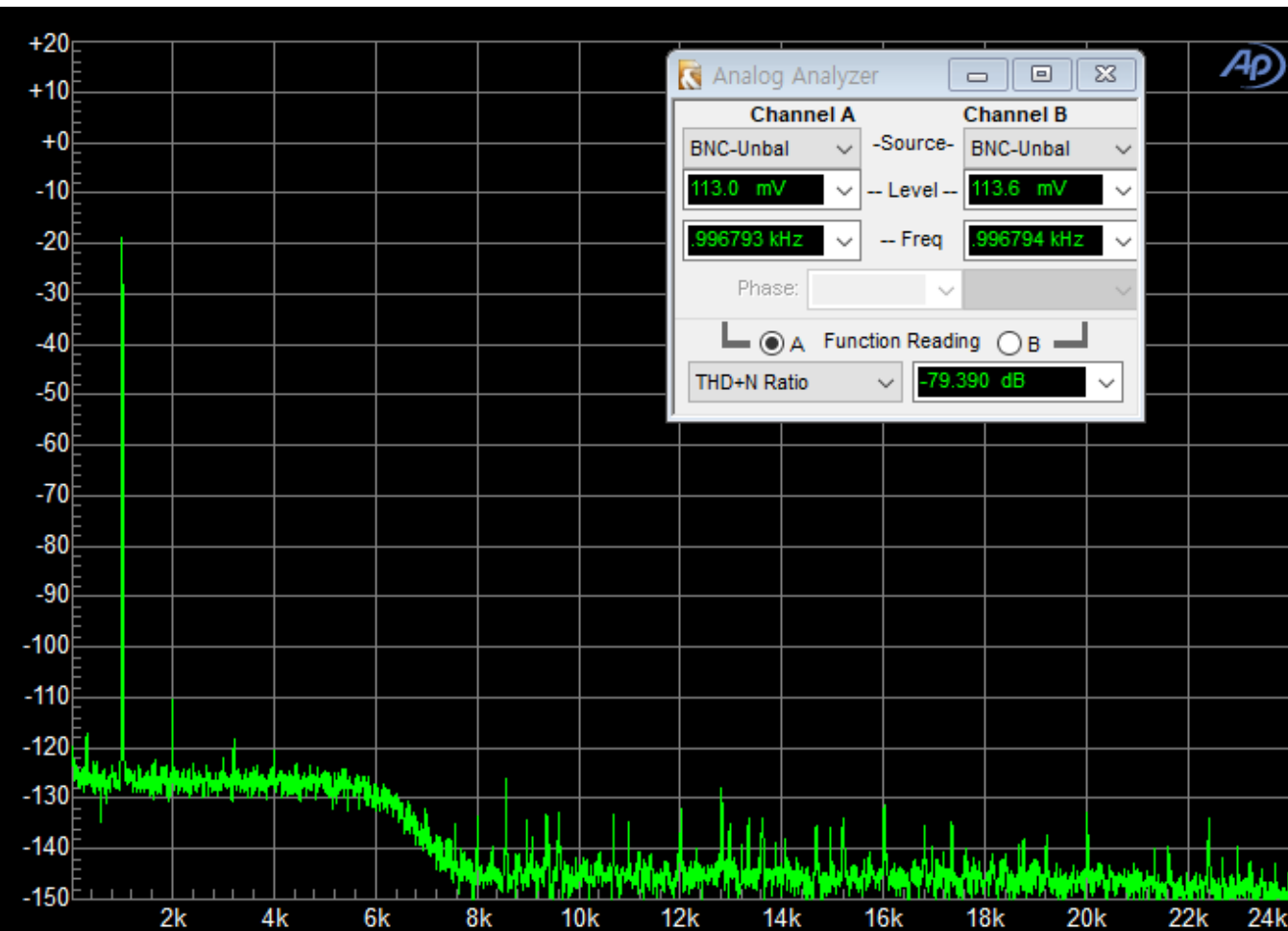
XB10 (aptX-HD)

Remark:

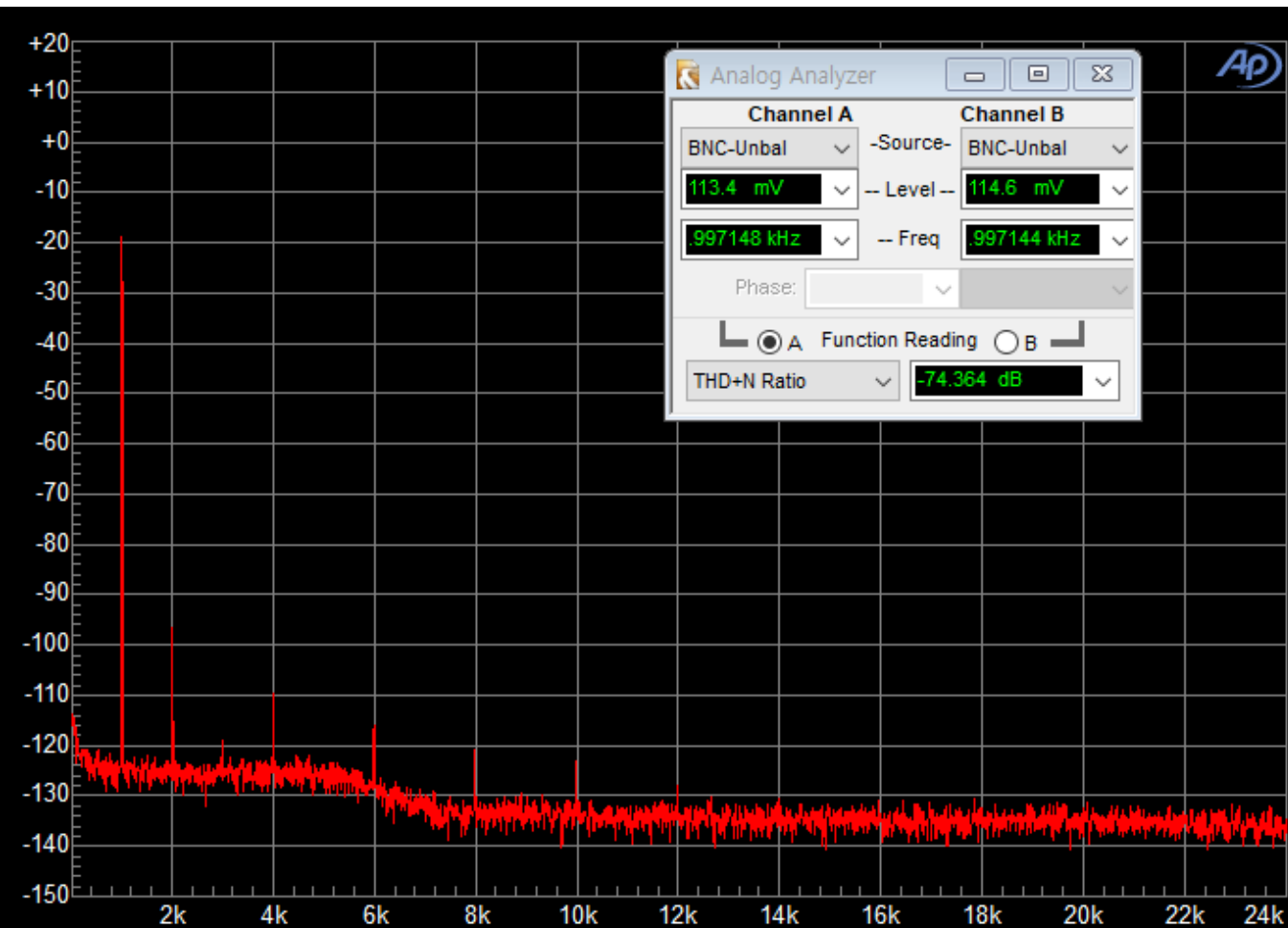
- When XBA-N1 (16Ω) load is connected, EarStudio (-79dB) is better than XB10 (-70dB) regarding THD+N at 360 mVrms output.
- The Noise Floor of EarStudio is lower than that of XB10. → Better SNR at the high-frequency region

SNR can be easily calculated from the differences between Peak and Noise Floor. AptX-HD codec characteristic determines the SNR below 6KHz, and both are the same. However, in the frequency range above 6KHz, you will find the differences in the Noise Floor, which eventually translates into SNR.

No Load @ 113 mVrms



EarStudio (aptX-HD)



XB10 (aptX-HD)

Remark:

- When XBA-N1 (16 Ω) load is connected, EarStudio (-79dB) is better than XB10 (-70dB) regarding THD+N at 113 mVrms output.
- The Noise Floor of EarStudio is lower than that of XB10. → Better SNR at high frequency region

EarStudio still delivers 79dB, which is almost same as the aptX-HD performance limitation, into 16 Ω loads across all volume levels.

2. USB DAC performance test

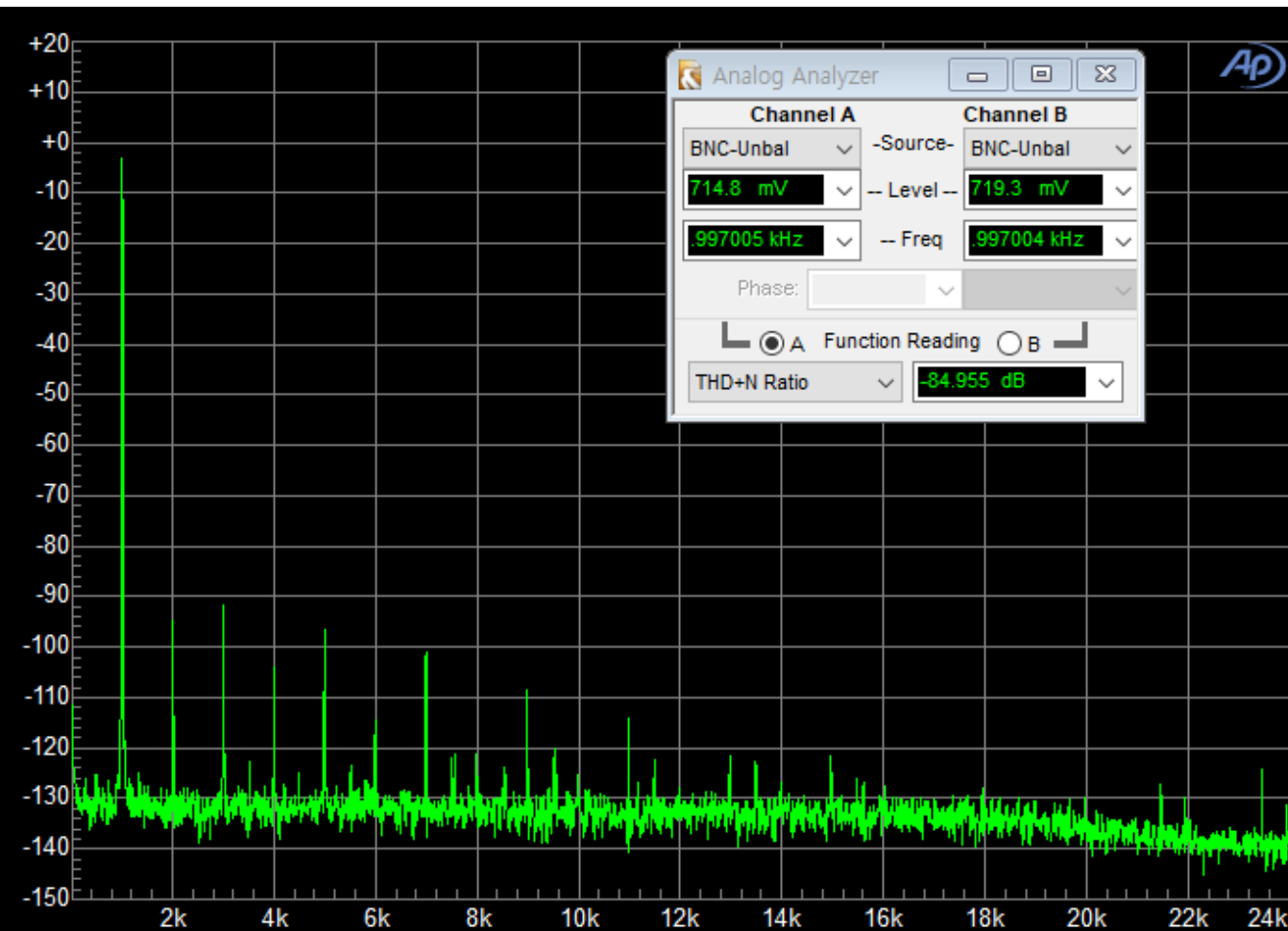
EarStudio vs. AK70

Test Setup:

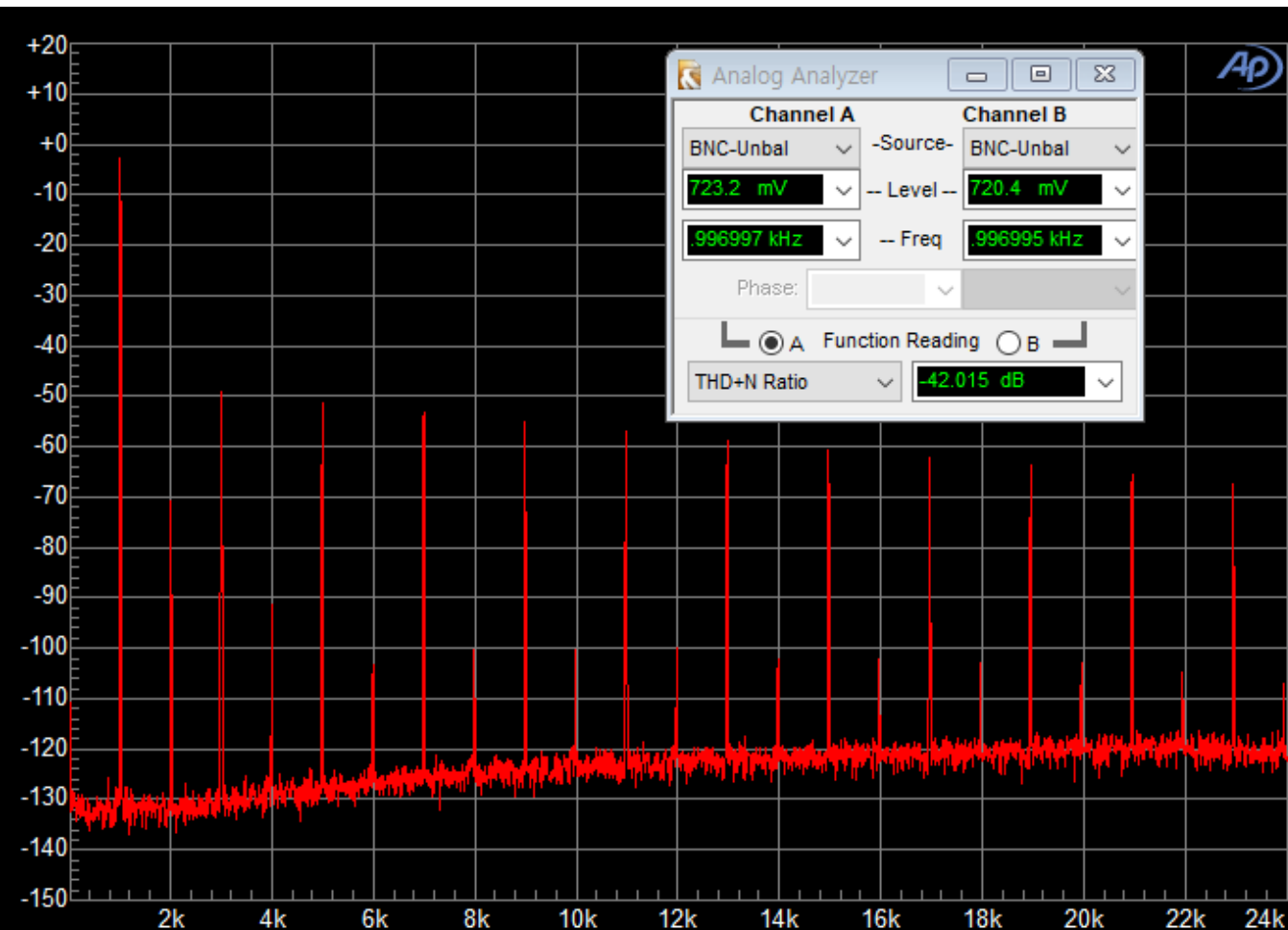
- EarStudio works as an USB DAC, PC plays and streams out the test tone.
- AK70 plays the local test tone.
- 997KHz test tone
- SONY XBA-N1 (16Ω) connected as a load (Standard 3.5mm cable)
- Measurement Equipment: AP2722
- DUT 3.5mm single-ended output to AP2722 unbalanced input



XBA-N1 (16Ω) @ 720 mVrms



EarStudio USB DAC



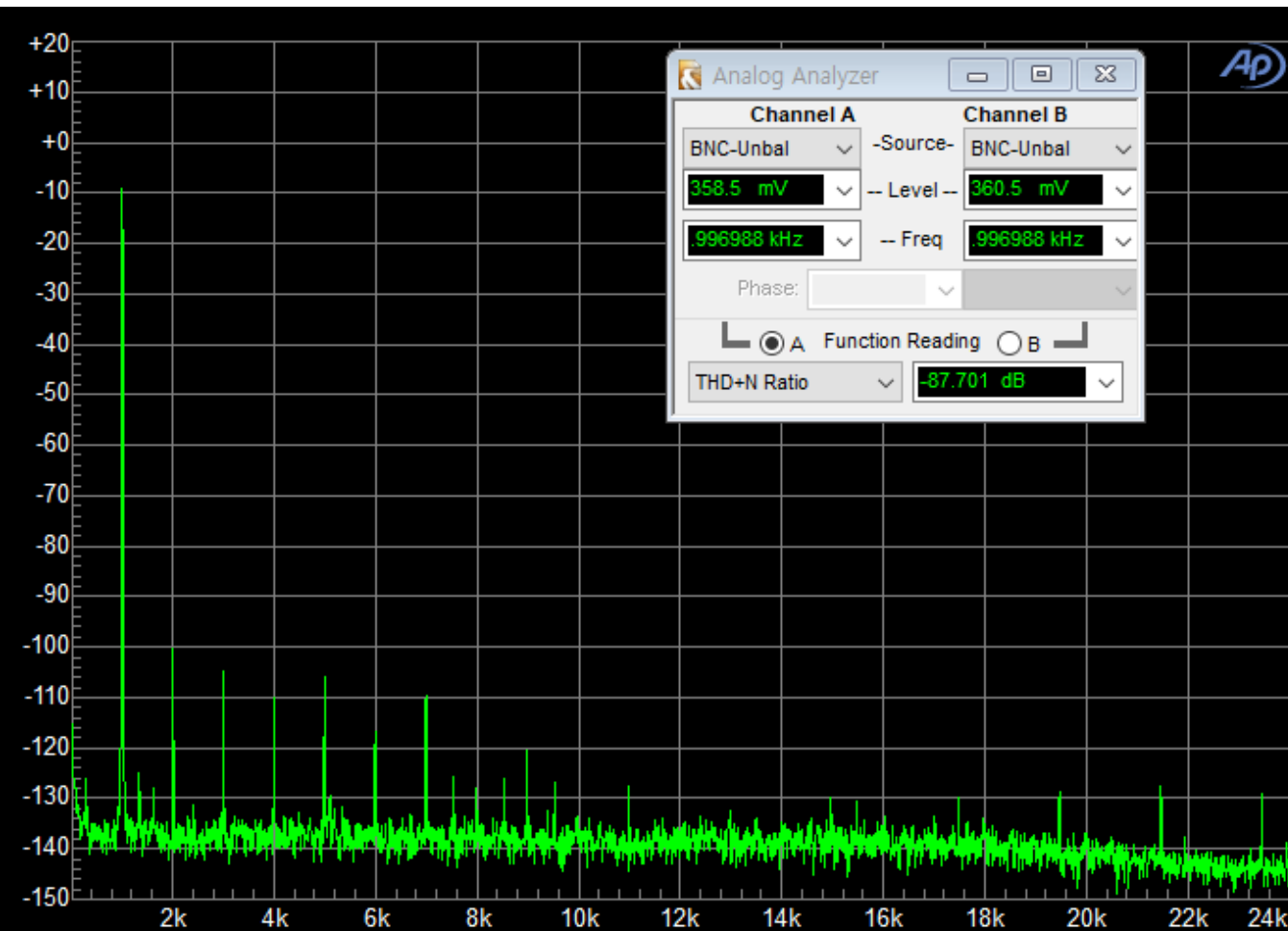
AK70 local file playback

Remark:

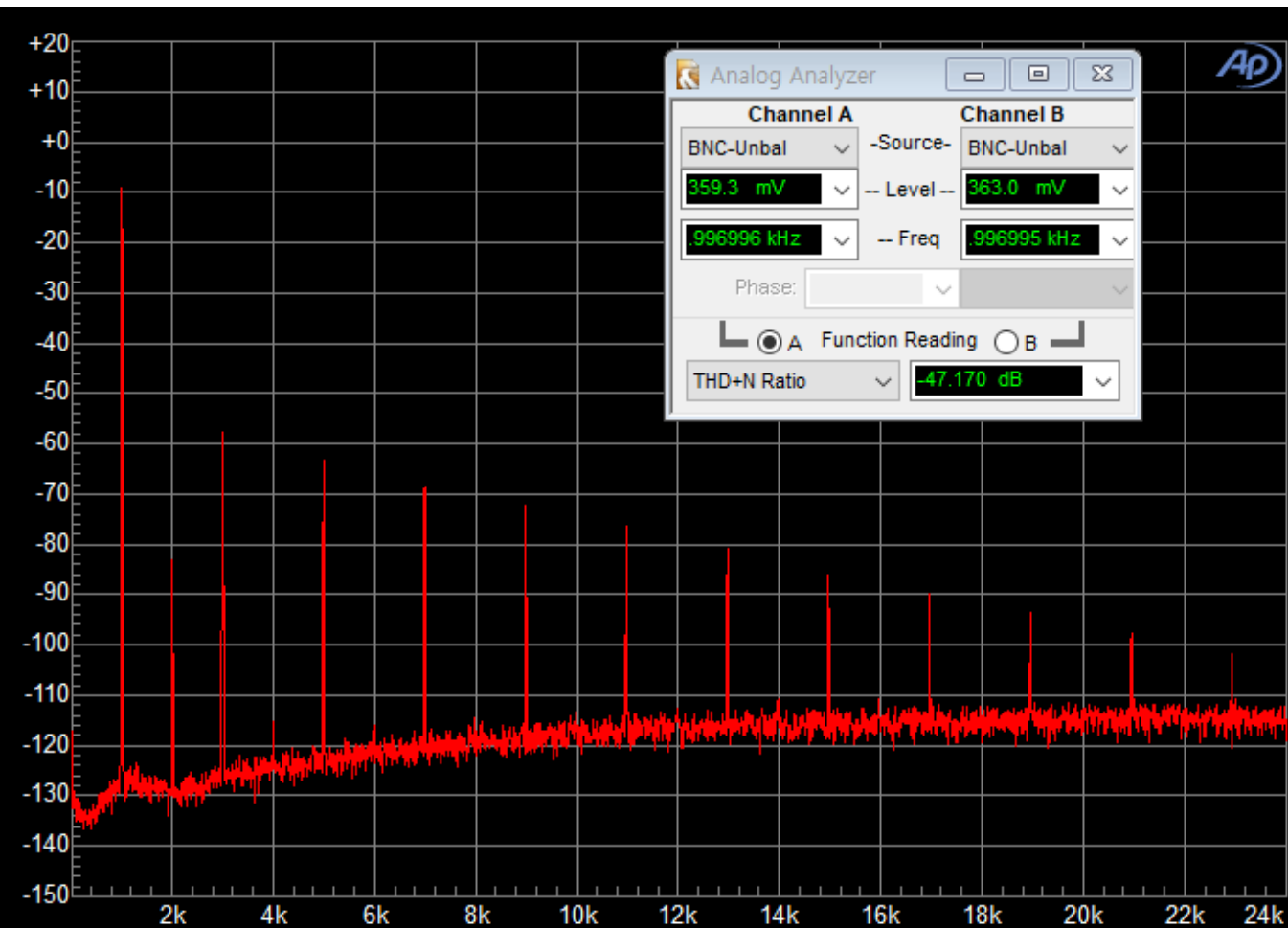
- When XBA-N1 (16Ω) load is connected, EarStudio (-84dB) is much better than AK70 (-42dB) regarding THD+N at 720 mVrms output.
- Approximately, 32.4mWatt output

16Ω Earphones are very much more sensitive to drive than high impedance headphones. It requires and consumes much more current flow from the analog output stage to the load, mostly resulting in considerable loss and distortion. The overall performances of the both devices get worse than the No load case. However, EarStudio shows the better THD+N than AK70.

XBA-N1 (16Ω) @ 360 mVrms



EarStudio USB DAC



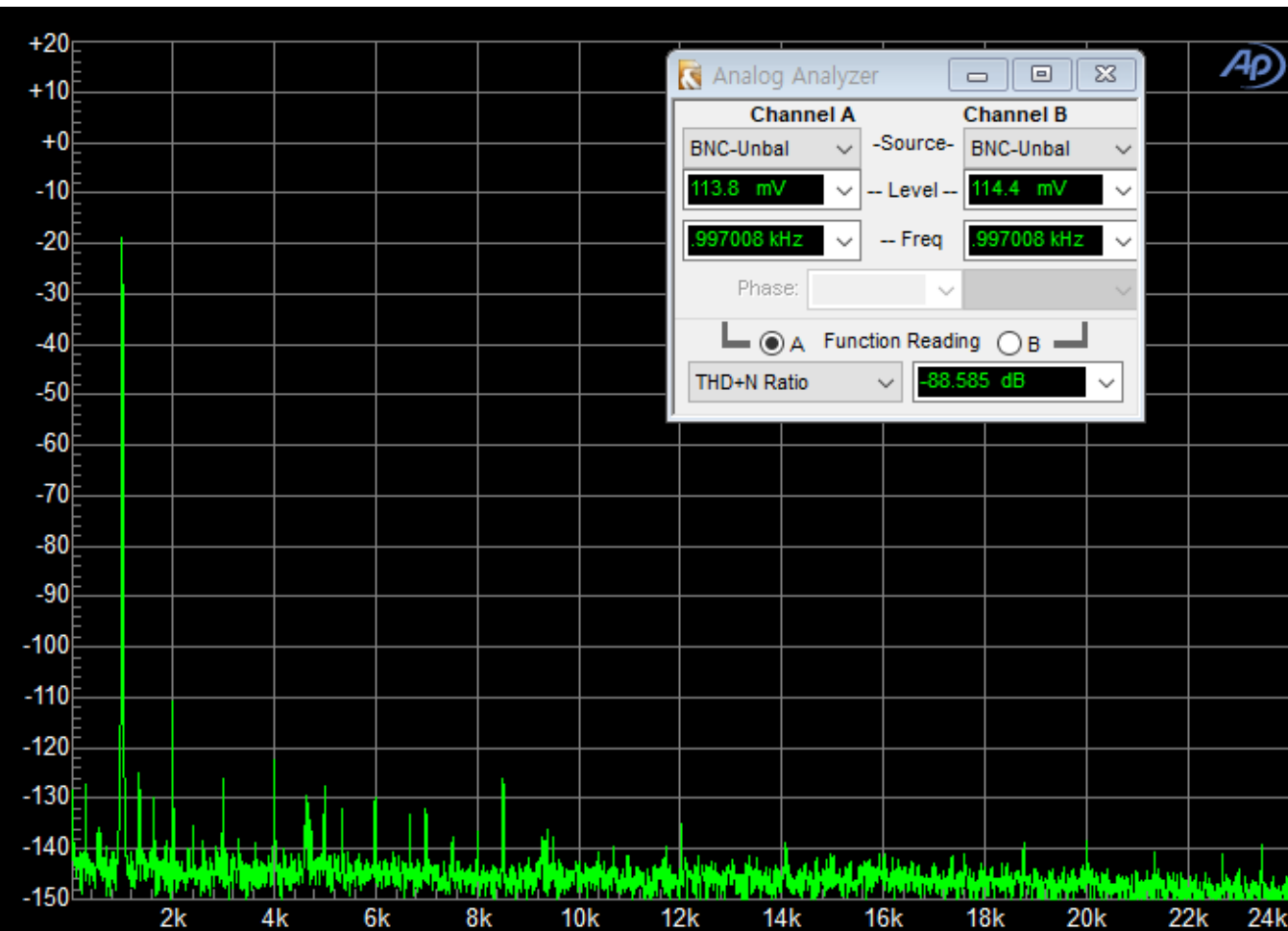
AK70 local file playback

Remark:

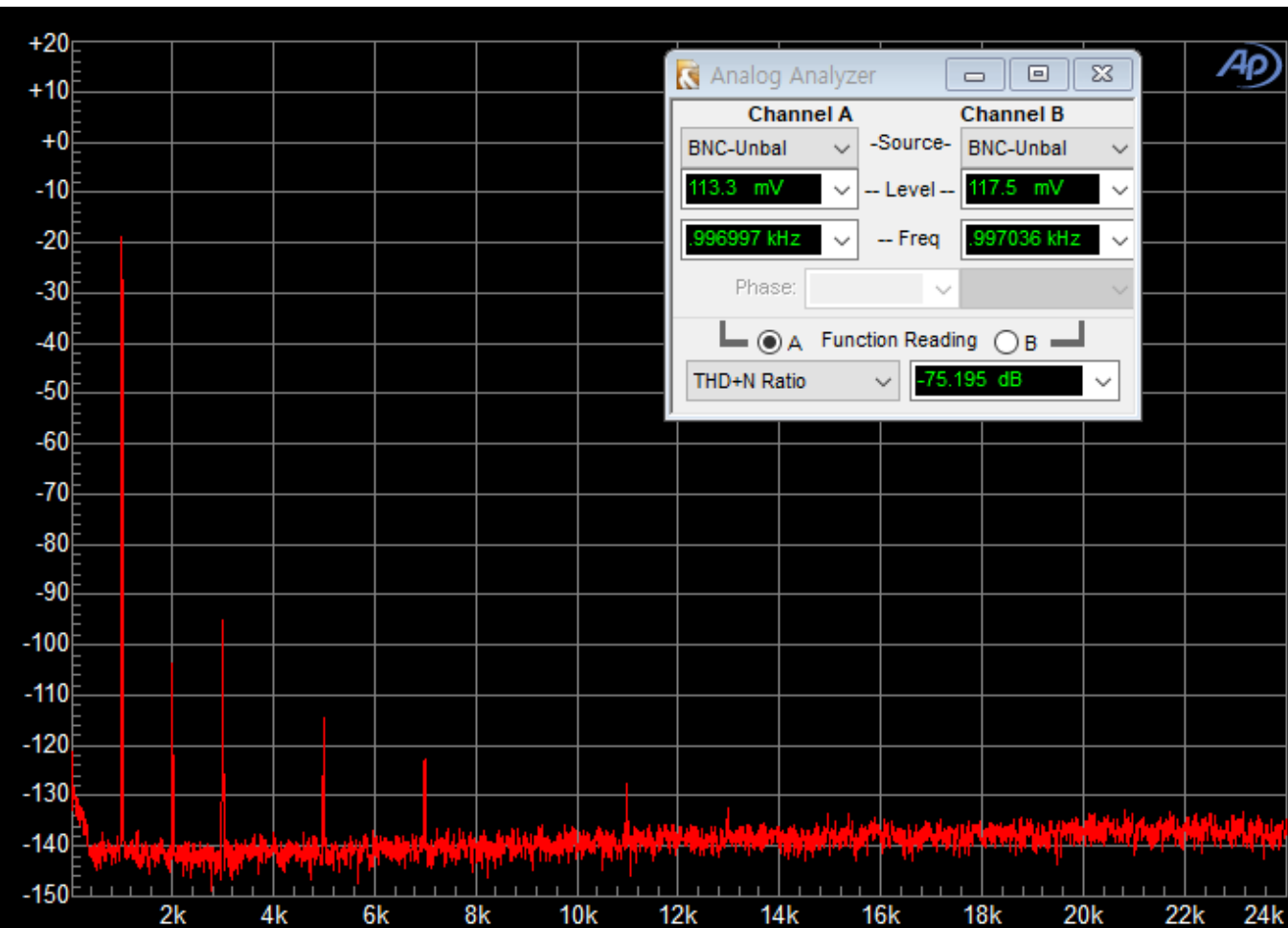
- When XBA-N1 (16Ω) load is connected, EarStudio (-87dB) is slightly better than AK70 (-47dB) regarding THD+N at 360 mVrms output.
- Approximately, 8.1mWatt output

As the output power decreased, the amount of current flow, distortion, and noise floor also get lowered. But still, AK70 shows relatively higher noise floor than EarStudio, as well as significantly odd harmonic distortion across all frequency range, which caused by overflow/saturation.

XBA-N1 (16Ω) @ 113 mVrms



EarStudio USB DAC



AK70 local file playback

Remark:

- When XBA-N1 (16Ω) load is connected, EarStudio (-88dB) is slightly better than AK70 (-75dB) regarding THD+N at 113 mVrms output.
- Approximately, 0.8mWatt output

Both results represent much better frequency response at low output power level. There is still more than 10dB performance difference even at low level between EarStudio and AK70.

Comments

In general, low impedance earphones are much more sensitive to drive than high impedance earphones. Especially, driving low impedance earphones requires much more current flow from the analog output stage to the load unit. Also, the device requires low output impedance. Otherwise, the current overflow at the amplifier or the voltage loss at the load would result in significant distortion and loss eventually.

Our proprietary DualDrive architecture can resolve those difficulties efficiently by using two DAC/AMP. For a single-ended output mode, DualDrive delivers twice much current at half output impedance than single DAC/AMP can do. The results of the test in this document demonstrate the advantage of the DualDrive architecture.

The test results show that:

As a universal headphone amplifier, DualDrive technology integrated into EarStudio offers the capability to drive any earphones or headphones ranged from 16Ω to 600Ω effortlessly. As an aptX-HD Bluetooth receiver, EarStudio delivers an outstanding performance at the analog output stage by reproducing the aptX-HD source without any additional distortion or loss in the analog domain.